Pradnya Sushil Shinde

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EDUCATION

Worcester Polytechnic Institute **Aug 2023 - May 2025**
*Master of Science in Robotics Engineering (**GPA: 4.0/4.0**)* *Worcester, MA*
Relevant Coursework: Computer Vision, Vision-based Robotic Manipulation, Motion Planning, Advanced Robot Navigation

Symbiosis Skills and Professional University **Aug 2019 - May 2023**
*Bachelor of Technology in Mechatronics Engineering (**GPA: 9.128/10.0**)* *Pune, India*

SKILLS

Languages: Python, C++, C
Libraries, Tools, and Software: Linux OS, ROS, ROS2, Nav2, Gazebo, Docker, RViz, OpenCV, PCL, SLAM Toolbox, NumPy, Matplotlib, Sklearn, PyTorch, Eigen, CUDA, Git, CMake, Blender, Arduino, MATLAB, Solidworks.
Hardware: Arduino, ESP32, ESP8266, Raspberry Pi, LiDAR

EXPERIENCE

Robotics Software Engineer **Jun '25 - Present**
Nightingale Intelligent Systems *Newark, CA*

- Co-own the onboard **ROS-based flight software stack** (**Python**) handling cloud-portal mission requests and drone state management; provisioned and deployed 20+ drones and base stations to production, resolving live field issues with customers.
- Built **FAA Remote ID integration** end-to-end: wrote a time-synchronized **ROS** publisher feeding the flight controller's Remote ID module, then cut onboard overhead via log filtering, achieving up to a **75**% reduction in message delays to meet the **3 s** compliance limit.
- Shipped flight features and reliability fixes across the stack, such as cloud-to-flight-controller speed control over **MAVLink**, concurrency bug fixes, and **ROS**-parameter runtime configuration.
- Extended and maintained the onboard C++ **path planner**, including terrain-aware altitude adjustment for site-specific AGL clearance, bug fixes, and capability improvements for planner robustness.

Robotics Software Engineer Intern **Jan '25 - Apr '25**
Nightingale Intelligent Systems *Newark, CA*

- Implemented **Multi-Heuristic A*** in **C++** for UAV path planning, pairing an admissible Euclidean anchor heuristic with weighted Manhattan, squared, and quartic heuristics to reduce node expansions compared to A* while preserving bounded suboptimality.
- Developed a simulation-only build of the **ROS** flight software that emulates onboard processing, enabling end-to-end testing without flight hardware.
- Deployed a Dockerized **ArduPilot SITL** environment (Gazebo server, GzWeb, ArduPilot SITL) on a shared server, used by the frontend team and for flight software validation.

Graduate Student Researcher **Aug '24 - May '25**
Worcester Polytechnic Institute *Remote*

- Evaluated Kimera, an open-source real-time metric-semantic **SLAM** library, for distributed mapping in agricultural robotics, building a data acquisition pipeline to benchmark its performance in field environments.

Robotics Software Engineer Intern **Aug '24 - Dec '24**
Celltrio Inc. *Fremont, CA*

- Performed kinematic and motion analysis on a **7-axis SCARA** robot to enable accurate tool transformations and end-effector control.
- Reviewed robot codebase to improve barcode scanning efficiency and motion performance, implementing code enhancements that increased operational efficiency.

Robotics Software Engineer Intern **May '24 - Aug '24**
National Robotics Engineering Center *Pittsburgh, PA*

- Integrated **SLAM Toolbox** into a **ROS 2** navigation stack for a wheeled warehouse mobile manipulator, enabling **LiDAR**-based mapping and localization for autonomous navigation.
- Validated sensor integration through unit testing and hardware-in-the-loop (**HIL**) testing on the physical robot.

Robotics Engineer Intern **Jun '22 - May '23**
Technodune Pvt. Ltd. *Pune, IN*

- Led software testing and embedded integration in **Python** and **C/C++** for a wirelessly controlled **7-DOF** bionic arm with gesture control and a web interface.
- Built a real-time robot localization platform using fiducial markers and **YOLOv5**, achieving **85**% pose detection accuracy.

PROJECTS

Einstein Vision: A Perception Pipeline for Autonomous Vehicles GitHub, Python

- Integrated lane detection, pedestrian pose estimation, road sign recognition, depth estimation, object detection, and scene classification into a unified perception stack, enabling seamless environmental visualization for an autonomous vehicle.

Motion and Trajectory Tracking of Aerial Vehicle GitHub, Python

- Implemented state estimation and global positioning algorithms such as **EKF**, **UKF**, **Particle Filter** and **GNSS/INS** to track the motion of an aerial vehicle effectively.

SfM: Structure from Motion GitHub, Python

- Engineered a robust a Structure from Motion (SfM) system to reconstruct 3D scenes and estimate camera poses from a monocular camera using images from multiple viewpoints.

3D Scene Reconstruction using NeRF GitHub, Python

- Developed a **NeRF**-based 3D scene reconstruction pipeline using a 5D coordinate system and **MLP** to optimize scene rendering from 2D images, and applied differentiable rendering to minimize residuals between synthesized and ground truth images.

Zhang's Camera Calibration: Optimizing Distortion Correction GitHub, Python

- Computed camera Intrinsic and Extrinsic Parameters by minimizing image distortion and reprojection error to perform Zhang's **Camera Calibration** method on checkerboard images.

Homography Estimation:Classical and Deep Leaning Methods GitHub, Python

- Designed a panorama stitching algorithm with corner detection, ANMS, feature matching, and **RANSAC**; trained CNN with TensorDLT and STN to predict homography, achieving $\mathbf{3.5-5.0}\%$ RMSE.

Probability-based Edge Detection: A Classical Approach to Boundary Detection GitHub, Python

- Developed a probability-based edge detection algorithm using filter banks, half-disc masks, k-means clustering for texture, brightness, and color features, and integrated with Sobel and Canny baselines for enhanced boundary detection.

Language Embodied Indoor Navigation GitHub, Python

- Developed a language-guided indoor navigation pipeline that converted spoken commands into navigation intents and executed goal-directed motion using **A*** and **RRT*** planners in Habitat, with RGB, depth, and semantic sensing for environment awareness.

Rigid Body Motion Planning GitHub, Python

- Developed and optimized **RRT**, **RRT***, **Informed RRT***, and **Probabilistic Roadmap**(PRM) based path planners tailored to a point robot, a 2D omnidirectional robot, and a 4-link kinematic chain.

Vision-based Object Grasping for Robotic Manipulation assuming Symmetry GitHub, C++

- Engineered a vision-based grasp pose analysis algorithm for robotic manipulation by processing depth point clouds, estimating symmetry for grasp point computation, using **Gazebo** and **ROS 2** for simulation.

PUBLICATION

Paper Title: "Marathi Sign Language Hand Gesture Recognition Using Accelerometer and 3D Printed Gloves" Paper
Status: Published in 2022 14th International Conference on Computational Intelligence and Communication Networks (CICN)
Publication Date: January 13, 2023